WEB PROGRAMMING MINI PROJECT

Course: Fundamentals of Computing (FOC)

Subject code – GXEST203 submitted

by

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To

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Practice Set -1

Q1: Create a simple div with an id "box". Add some text content inside the div. Set its background colour to blue.

Q2: Create 3 headings with h1, h2 & h3. Give them all a class "heading" & set colour of "heading" to red.

Q3: Create a button & set its background colour to:

- green using css stylesheet
- blue using <style> tag • pink using inline code

Source Code

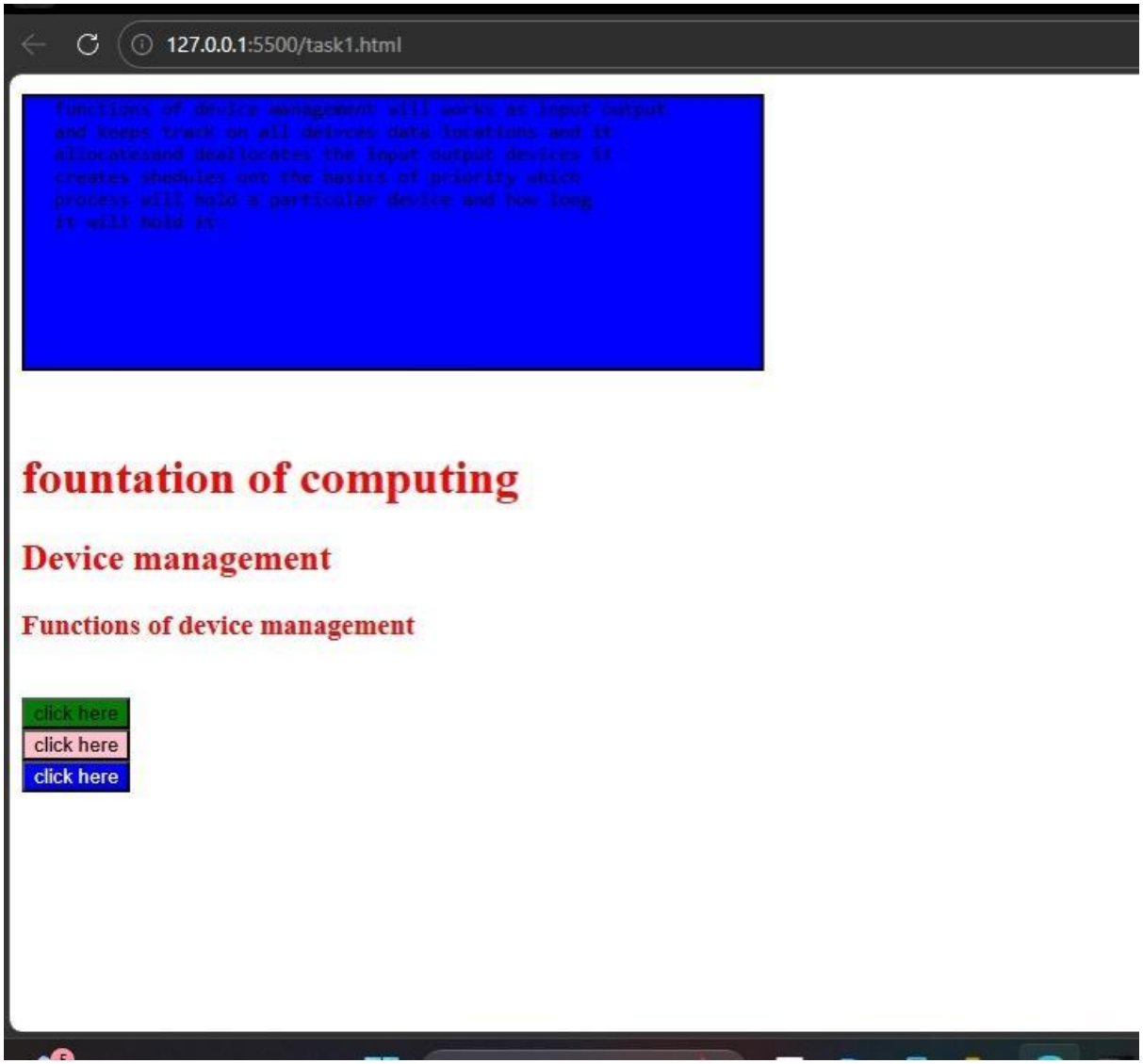
HTML**:**

```
Go  Run  ···                ccf
html task1.html ×   task6.html U   task6.css U   task7.html U   task8.html U
C: > Users > HP > OneDrive > Documents > html task1.html > html > body > div > h2.headin
<html lang="en">
<head>
    <style>
        .button{
        }
    </style>
</head>
<body>
    <div id="box">
        <pre id="id1">
functions of device management will works as input output
and keeps track on all deivces data locations and it
allocatesand deallocates the input output devices it
creates shedules ont the basics of priority which
process will hold a particular device and how long
it will hold it.
</pre><br>
</div> <div>
<h1 class ="heading">fountation  of computing</h1>
<h2 class="heading">Device management</h2>
<h3 class="heading">Functions of device management</h3>
</div><br>
<button id="id2">click here</button><br>
<button style ="background-color: pink;">click here</button><br>
<button class="button">click here</button>
</body>
</html>
```

CSS:

```
# task1.css > .button
#id1{
    background-color: blue;
    height: 180px;
    width: 450px;
    font-weight: 500;
    border: 2px solid black;
    padding-left: 20px;
    padding-right: 20px;
}
.heading{
    color: red;
}
#id2{
    background-color: green;
}
.button{
    background-color: blue;
    color: white;
}
```

Output:


127.0.0.1:5500/task1.html
Functions of device management will works as input output
and keeps track on all devices data locations and it
allocatesand deallocates the input output devices it
creates sheduler ont the basics of priority which
process will hold a particular device and how long
it will hold it.
fountation of computing
Device management
Functions of device management
click here
click here
click here

Practice Set -2

Q1: Create a heading centred on the page with all of its text capitalized by default.

Q2: Set the font family of all the content in the document to "Times New Roman".

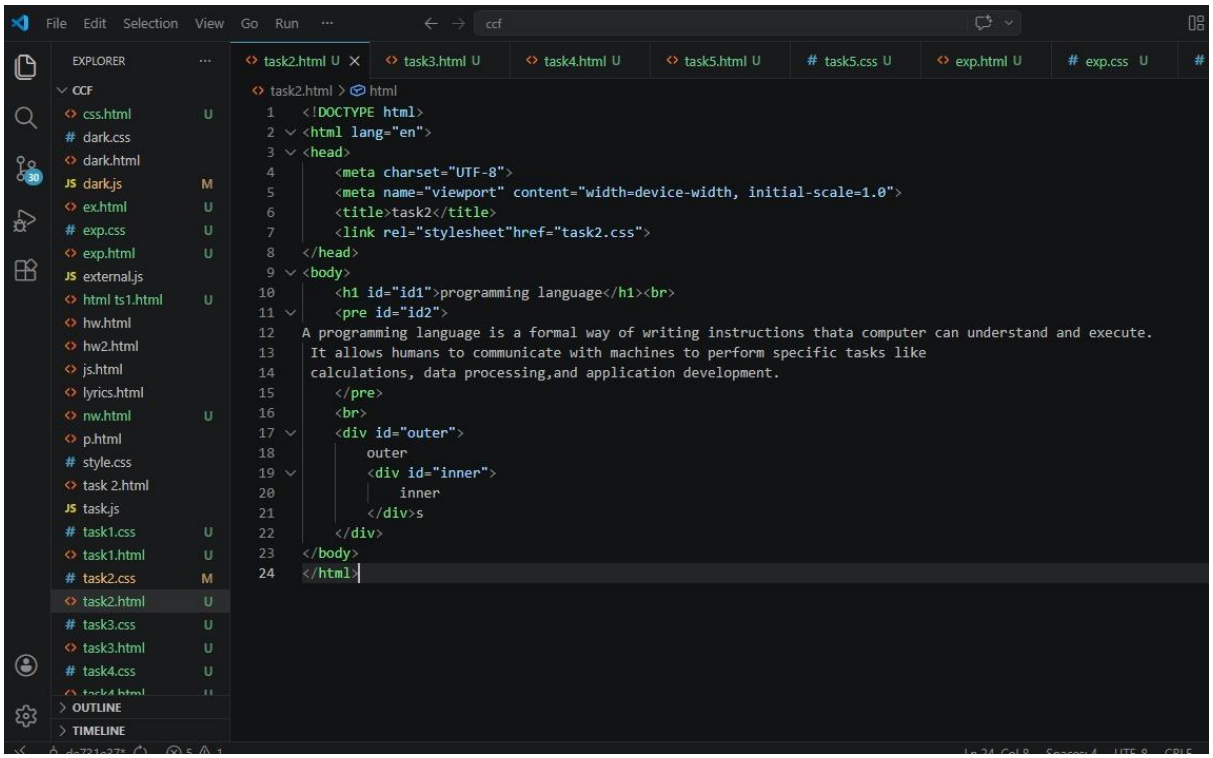
Q3: Create one div inside another div.

Set id & text "outer" for the first one & "inner" for the second one.

Set the outer div text size to 25px & inner div text size to 10px.

Source Code

HTML:

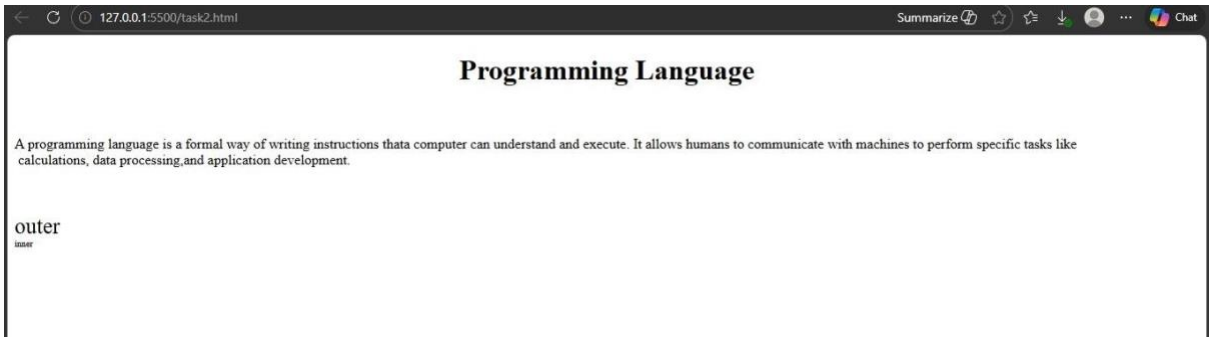


```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>task2</title>
    <link rel="stylesheet"href="task2.css">
</head>
<body>
    <h1 id="id1">programming language</h1><br>
    <pre id="id2">
 A programming language is a formal way of writing instructions thata computer can understand and execute.
  It allows humans to communicate with machines to perform specific tasks like
  calculations, data processing,and application development.
    </pre>
    <br>
    <div id="outer">
        outer
        <div id="inner">
            inner
        </div>s
    </div>
</body>
</html>
```

CSS:

```
#id1{
    text-align: center;
    text-transform: capitalize;
}
#id2{
    font-family: "Times New Roman";
}
#outer{
    font-size: 25px;
}
#inner{
    font-size: 10px;
}
```

Output:



Practice Set - 3

Q1: Create a div with height & width of 100px. Set its background colour to green & the border radius to 50%.

Q2: Create the following navbar.

Source Code

HTML:

```
Go  Run  ···                ccf
ml U   <> task4.html U   <> task5.html U
# task3.css > .search button
}

/* Links */
.links a {
    color: white;
    text-decoration: none;
    margin-right: 20px;
}

/* Search section */
.search {
    margin-left: auto;
}

.search input {
    padding: 5px;
}

.search button {
    background-color: #f08804;
    border: none;
    padding: 6px 10px;
    color: white;
}
```

CSS :-

```html
<!DOCTYPE html>
<html>
<head>
    <link rel="stylesheet" href="task3.css">
</head>
<body>
<br>
<div id="circle"></div> <br>
<div class="navbar">

    <div class="logo">amazon.in</div>

    <div class="links">
        <a href="#">Account</a>
        <a href="#">My Cart</a>
        <a href="#">Contact Us</a>
    </div>

    <div class="search">
        <input type="text" placeholder="search Amazon.in">
        <button>Search</button>
    </div>

</div>
</body>
</html>
```

Output:

Practice Set -4

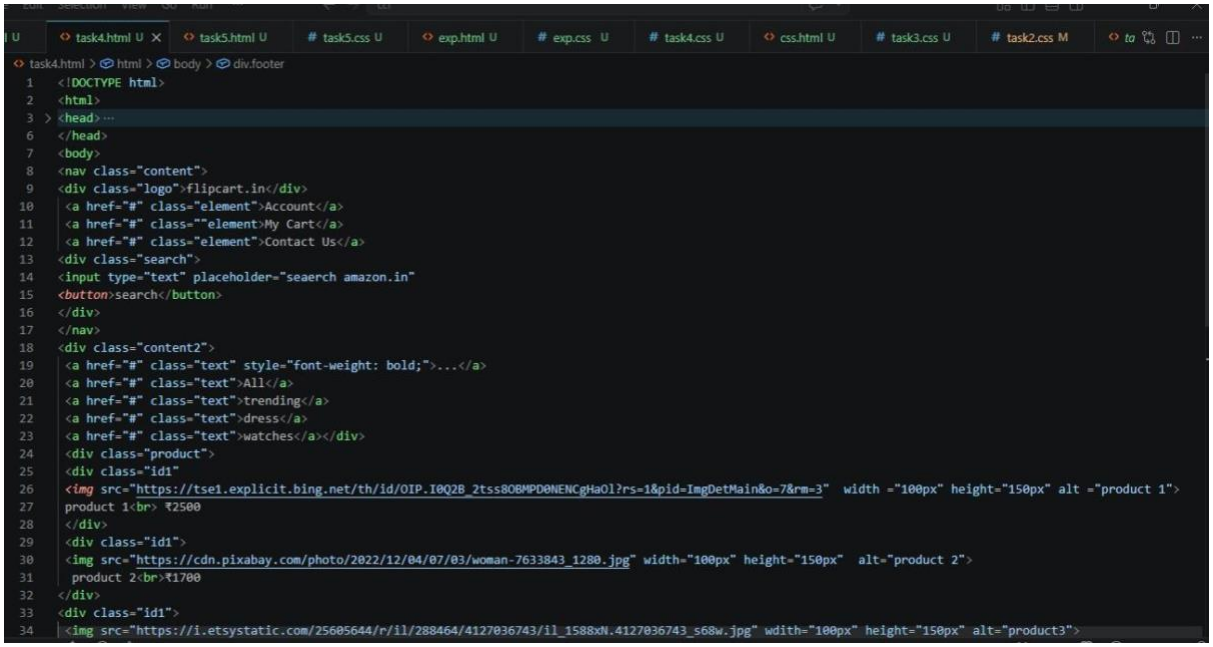
Q1: Create a webpage layout with a header, a footer & a content area containing 3 divs. Set the height & width of divs to 100px. (add the previous navbar in the header) Q2: Add borders to all the divs.

Q3: Add a different background colour to each div with an opacity of 0.5.

Q4: Give the content area an appropriate height

Source code

HTML:



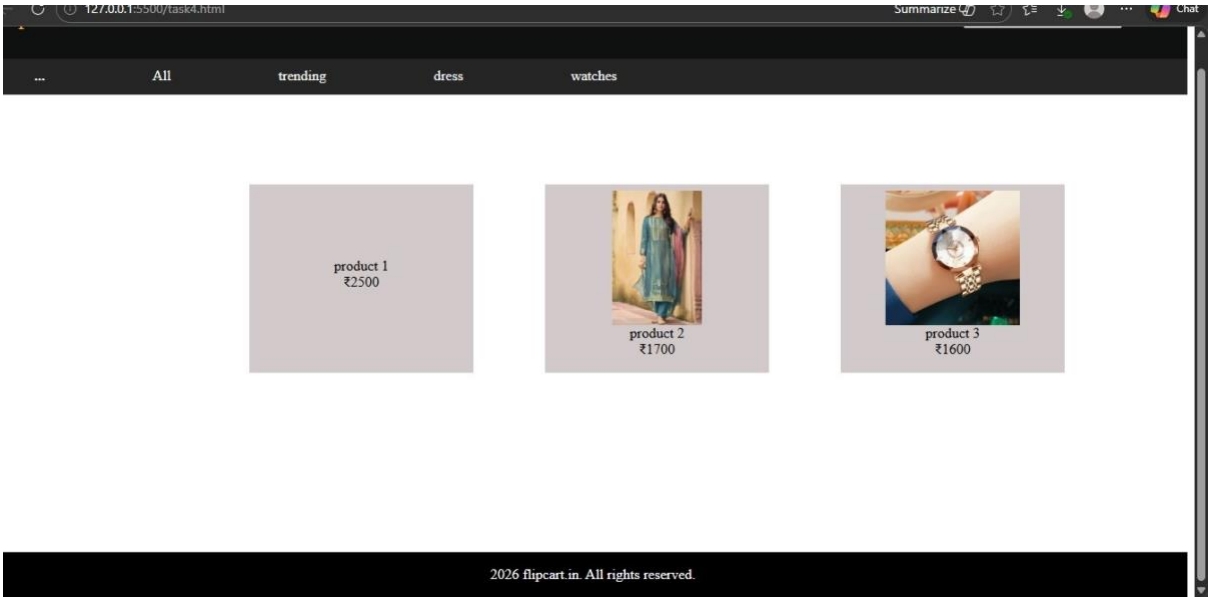
```
<!DOCTYPE html>
<html>
<head>···
</head>
<body>
<nav class="content">
<div class="logo">flipcart.in</div>
 <a href="#" class="element">Account</a>
 <a href="#" class=""element>My Cart</a>
 <a href="#" class="element">Contact Us</a>
<div class="search">
<input type="text" placeholder="seaerch amazon.in"
<button>search</button>
</div>
</nav>
<div class="content2">
 <a href="#" class="text" style="font-weight: bold;">...</a>
 <a href="#" class="text">All</a>
 <a href="#" class="text">trending</a>
 <a href="#" class="text">dress</a>
 <a href="#" class="text">watches</a></div>
 <div class="product">
 <div class="id1"
 <img src="https://tse1.explicit.bing.net/th/id/OIP.I0Q2B_2tss8OBMPD0NENCgHaOl?rs=1&pid=ImgDetMain&o=7&rm=3"  width ="100px" height="150px" alt ="product 1">
 product 1<br> ₹2500
 </div>
 <div class="id1">
 <img src="https://cdn.pixabay.com/photo/2022/12/04/07/03/woman-7633843_1280.jpg" width="100px" height="150px"  alt="product 2">
  product 2<br>₹1700
</div>
<div class="id1">
 <img src="https://i.etsystatic.com/25605644/r/il/288464/4127036743/il_1588xN.4127036743_s68w.jpg" wdith="100px" height="150px" alt="product3">
```

```html
<html>
<body>
<div class="product">
<div class="id1"
<img src="https://tse1.explicit.bing.net/th/id/OIP.I0Q2B_2tss8OBMPD0NENCgHaOl?rs=1&pid=ImgDetMain&o=7&rm=3"  width ="100px" height="150px" alt ="product 1">
product 1<br> ₹2500
</div>
<div class="id1">
<img src="https://cdn.pixabay.com/photo/2022/12/04/07/03/woman-7633843_1280.jpg" width="100px" height="150px"  alt="product 2">
product 2<br>₹1700
</div>
<div class="id1">
<img src="https://i.etsystatic.com/25605644/r/il/288464/4127036743/il_1588xN.4127036743_s68w.jpg" wdith="100px" height="150px" alt="product3">
product 3<br>₹1600
</div>
</div><div class="footer">
<p>2026 flipcart.in. All rights reserved.</p>
</div>

</body>
</html>
```

CSS:-

```css

.product {
    flex-direction: row;
    display: flex;
    gap:80px;
    margin-left:280px;
    padding-top: 100px;
}
.footer {
    background-color: black;
    color: white;
    height: 50px;
    display: flex;
    justify-content: center;
    align-items: center;
    margin-top: 200px;
}
```

Output:



Practice Set -5

Qs: Create the following layout using the given html.

- Give the div a height, width & some background image.
-
- Use the appropriate position property for the div element to place it at the right end of the page. (The div should not move even on scroll)
-
- Use z-index to place the div on top of page.
-

lorem*5

Love Nature

lorem*5</p

Sourcecode:-

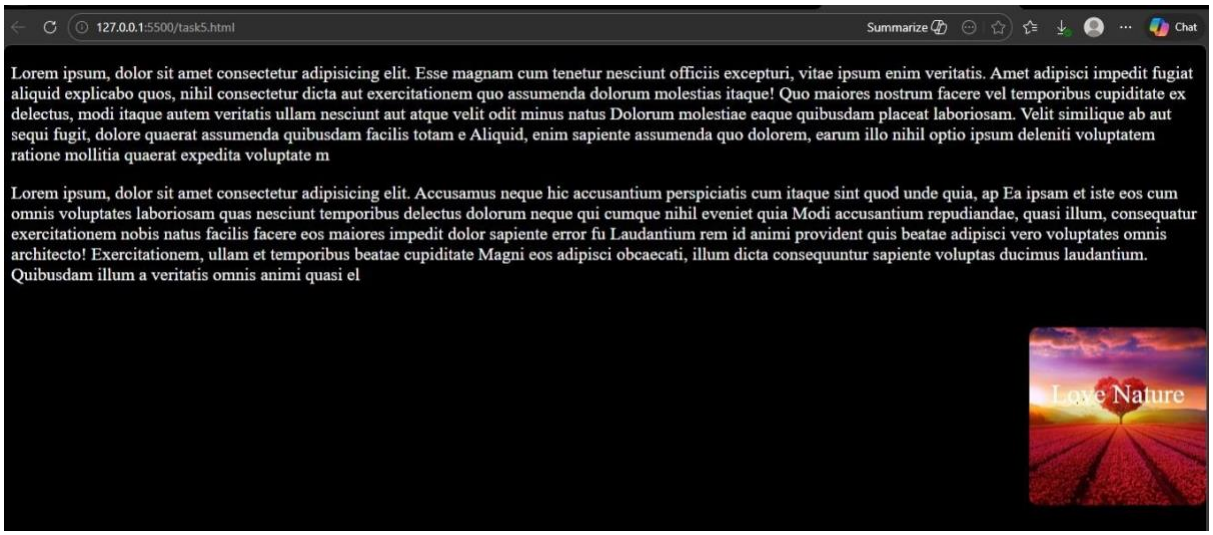
HTML:-

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>task5</title>
    <link rel="stylesheet" href="task5.css">
</head>
<body>

    <p>
  Lorem ipsum, dolor sit amet consectetur adipisicing elit. Esse magnam cum tenetur nesciunt officiis excepturi, vitae ipsum enim veritatis.
  Amet adipisci impedit fugiat aliquid explicabo quos, nihil consectetur dicta aut exercitationem quo assumenda dolorum molestias itaque!
  Quo maiores nostrum facere vel temporibus cupiditate ex delectus, modi itaque autem veritatis ullam nesciunt aut atque velit odit minus natus
  Dolorum molestiae eaque quibusdam placeat laboriosam. Velit similique ab aut sequi fugit, dolore quaerat assumenda quibusdam facilis totam e
  Aliquid, enim sapiente assumenda quo dolorem, earum illo nihil optio ipsum deleniti voluptatem ratione mollitia quaerat expedita voluptate m
</p>

<div class="box">Love Nature</div>

<p>
  Lorem ipsum, dolor sit amet consectetur adipisicing elit. Accusamus neque hic accusantium perspiciatis cum itaque sint quod unde quia, ap
  Ea ipsam et iste eos cum omnis voluptates laboriosam quas nesciunt temporibus delectus dolorum neque qui cumque nihil eveniet quia
  Modi accusantium repudiandae, quasi illum, consequatur exercitationem nobis natus facilis facere eos maiores impedit dolor sapiente error fu
  Laudantium rem id animi provident quis beatae adipisci vero voluptates omnis architecto! Exercitationem, ullam et temporibus beatae cupiditate
  Magni eos adipisci obcaecati, illum dicta consequuntur sapiente voluptas ducimus laudantium. Quibusdam illum a veritatis omnis animi quasi el
</p>
</body>
</html>
```

CSS:-

```css
body {
    font-family: 'Times New Roman', Times, serif;
    font-size: 20px;
    background-color: black;
    color: white;
}


.box {
    height: 200px;
    width: 200px;
    background-image: url(https://wallpapers.com/images/featured/love-nature-qdl3kg0gtps6l9c3.jpg);
    background-size: cover;
    background-position: center;
    color: white;
    font-size: 30px;
     text-align: center;
     line-height: 150px;
    position: fixed;
    right: 0;
    top: 50%;
    z-index: 1000;
    border-radius: 10px;
}
```

OUTPUT:-



Practice Set -6

Q1: Create a navbar with 4 options in the form of anchor tags inside list items. Now, use flexbox to place them all spaced equally in a single line.

Q2: Use flexbox to centre one div inside another div.

Q3: Which has higher priority - align-items or align-self?

Answer: align-self has higher priority than align-items.

Source code HTML**:**

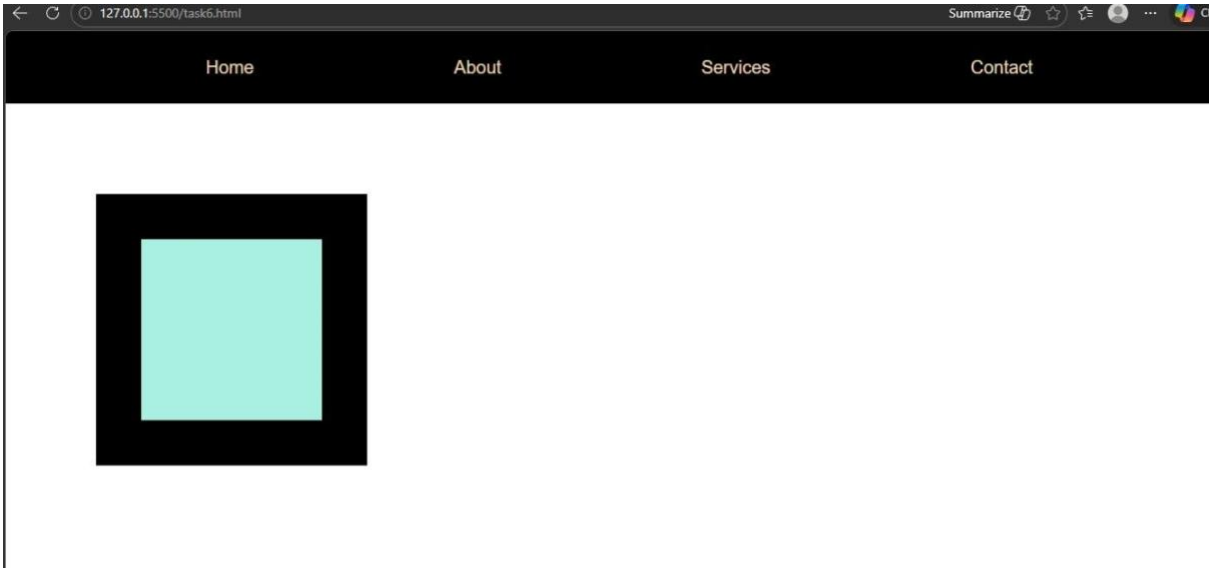
```html
<!DOCTYPE html>
<html>
<head>
    <title>task6</title>
    <link rel="stylesheet" href="task6.css">
</head>
<body>
<div>

    <nav class="navbar">
        <div><a href="#" class="element">Home</a></div>
        <div><a href="#" class="element">About</a></div>
        <div><a href="#" class="element">Services</a></div>
        <div><a href="#" class="element">Contact</a></div>
    </nav>
</div>
    <div class="center">
        <div class="inner"></div>
    </div>


</body>
</html>
```

CSS:

```
6.html U    # task6.css U ×    task7.html U
k6.css > .center
body {
    margin: 0;
    font-family: Arial;
}

.navbar {
    display: flex;
    justify-content: space-evenly;
    flex-direction: row;
    align-items: center;
    color: white;
    background-color: black;
    height: 80px;
}
.element{
    font-size: 20px;
    text-decoration: none;
    color: wheat;
}
.center{
    display: flex;
    justify-content: row;
    align-items: center;
    height: 300px;
    width: 300px;
    margin: 100px;
    background-color: black;
    justify-content: center;
}
.inner{
    height: 200px;
    width: 200px;
37*  ⊗ 5 ⚠ 1
```

```
sk6.html U    # task6.css U ×    task7.html U
ask6.css > .center
}
.element{
    font-size: 20px;
    text-decoration: none;
    color: wheat;
}
.center{
    display: flex;
    justify-content: row;
    align-items: center;
    height: 300px;
    width: 300px;
    margin: 100px;
    background-color: black;
    justify-content: center;
}
.inner{
    height: 200px;
    width: 200px;
    background-color: rgb(170,240, 224);

}
```

Output:



Practice Set -7

Qs: Add a media query to implement the following:

- The colour of a div changes to green for viewport width less than 300px.
- The colour of a div changes to pink for width between 300px & 400px.
- The colour of a div changes to red for width between 400px & 600px.
- The colour of a div changes to blue for width above 600px. Source

SOURCECODE:-

HTML:-

ew Go Run ··· ccf

task6.html U | task6.css U | task7.html U X | task8.html U | task9.html U | task1

task7.html > html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Media Query Example</title>
    <link rel="stylesheet" href="task7.css">
</head>
<body>

    <div class="box">
</body>
</html>
```

CSS:-

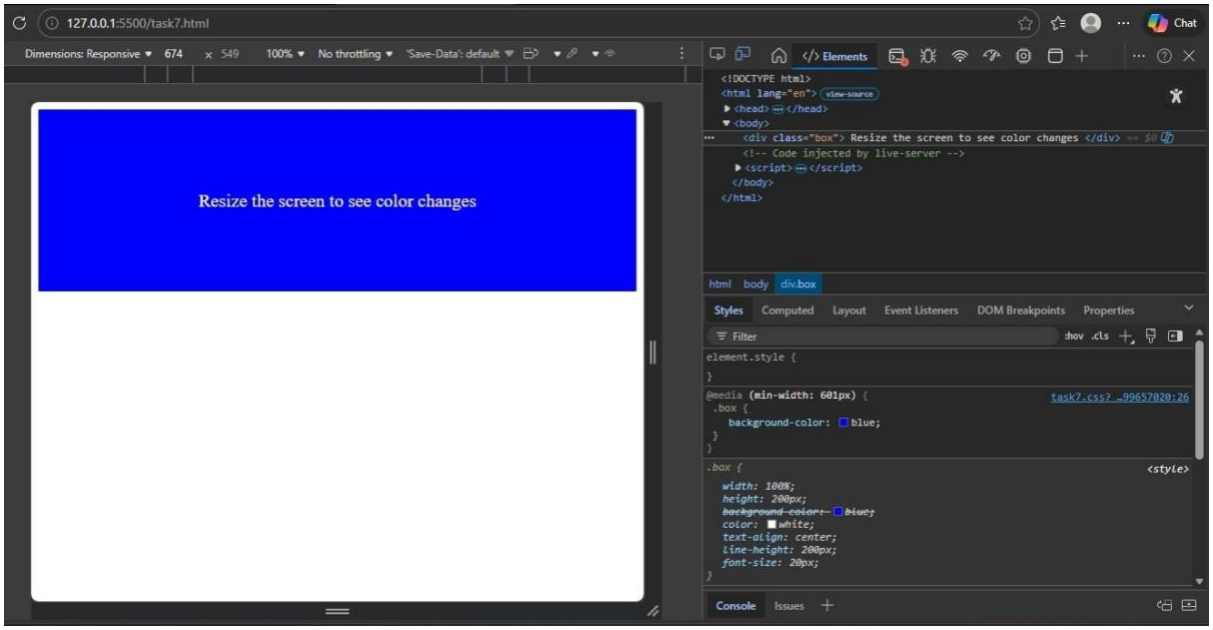
Go Run ··· ccf

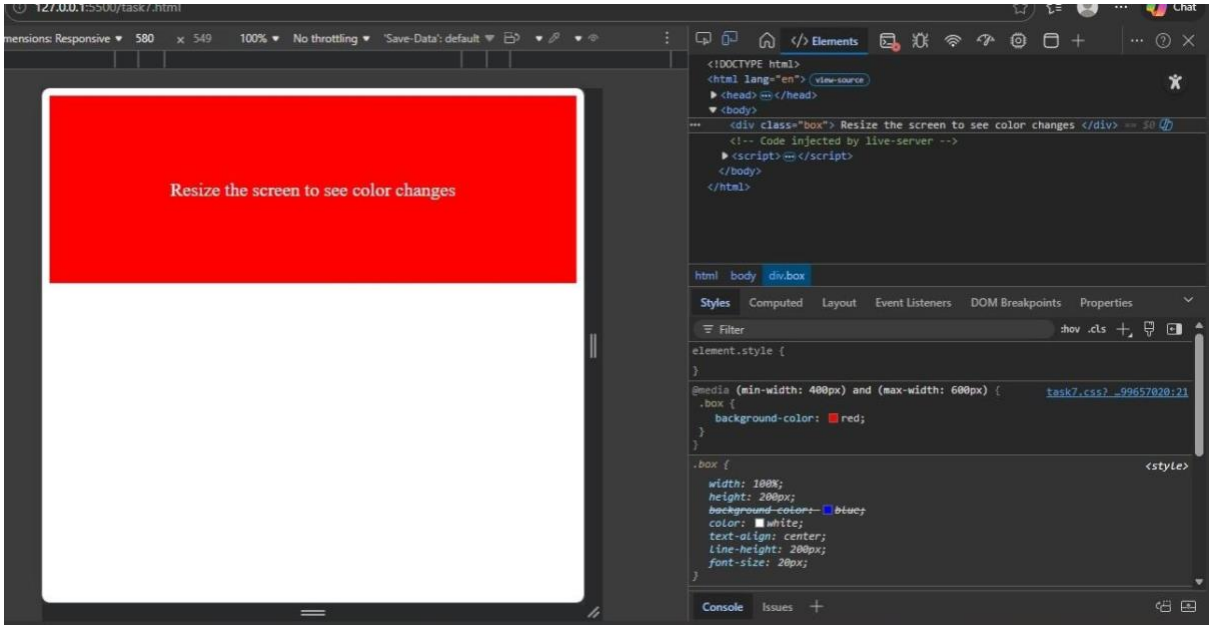
task10.html U | task10.js U | task11.html U | task11.css 6, U

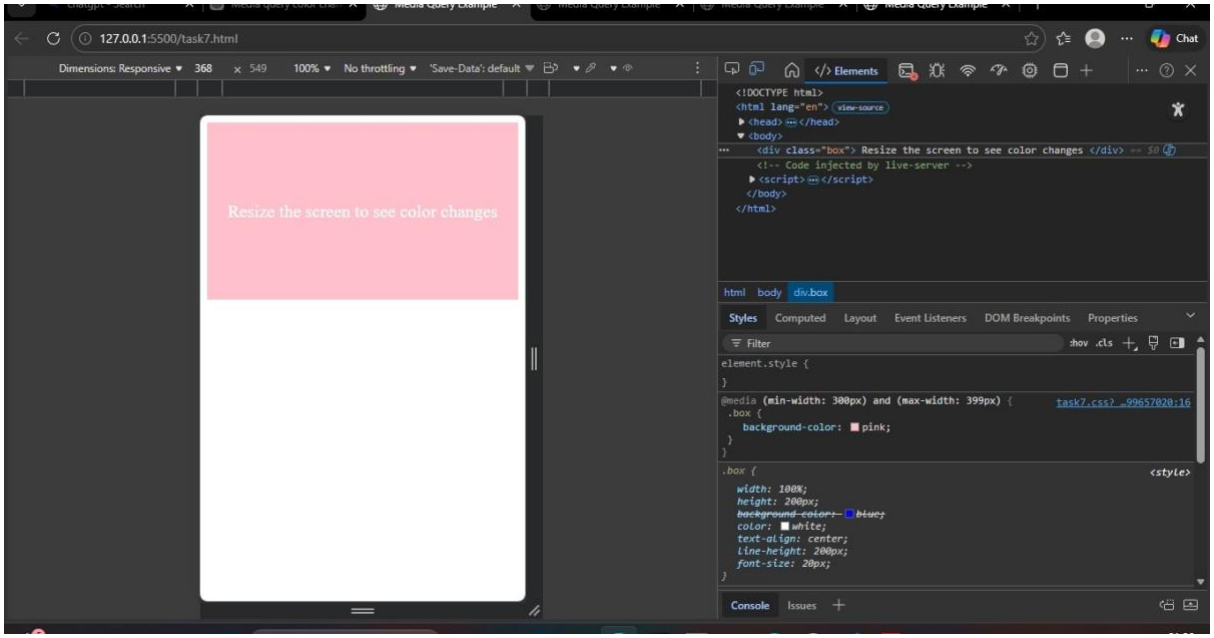
task7.css > { } @media (min-width: 601px)

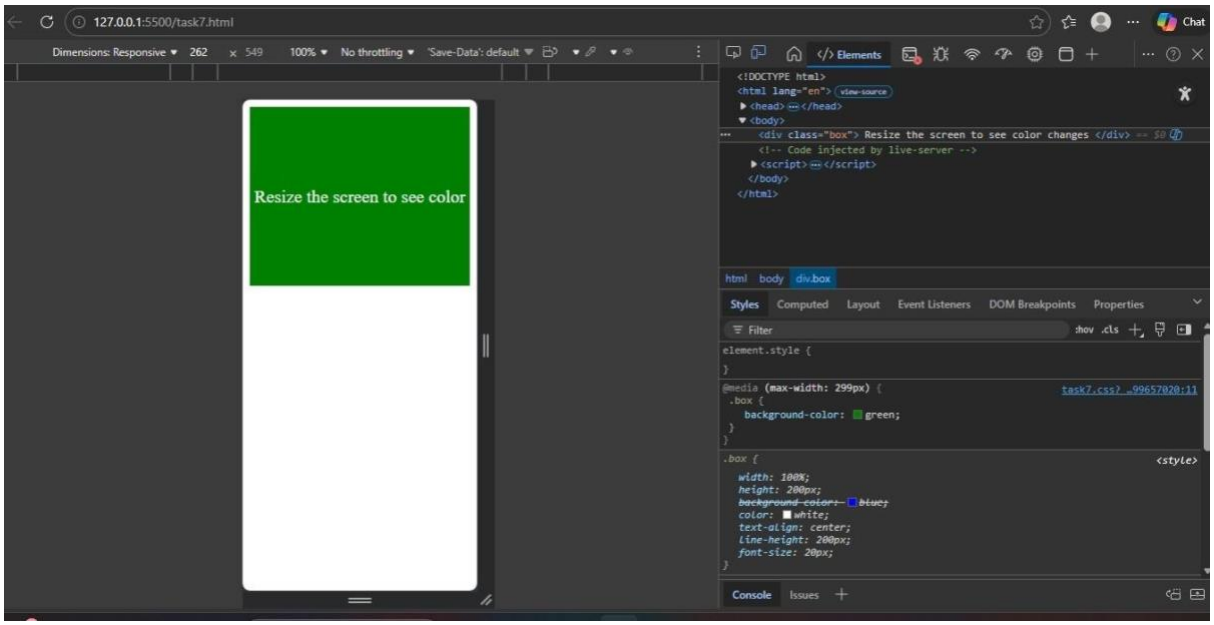
```
.box {
    height: 200px;
    display: flex;
    flex: auto;
    justify-content: center;
    align-items: center;
}
@media (max-width: 299px) {
    .box {
        background-color: green;
    }
}
@media (min-width: 300px) and (max-width: 399px) {
    .box {
        background-color: pink;
    }
}
@media (min-width: 400px) and (max-width: 600px) {
    .box {
        background-color: red;
    }
}
@media (min-width: 601px) {
    .box {
        background-color: blue;
    }
}
```

OUTPUT:-


127.0.0.1:5500/task7.html
Dimensions: Responsive 674 x 549 100% No throttling 'Save-Data': default
Resize the screen to see color changes
Elements
<!DOCTYPE html>
<html lang="en">
<head>
<body>
Resize the screen to see color changes
<!-- Code injected by live-server -->
<script>
</body>
</html>
html body div.box
Styles Computed Layout Event Listeners DOM Breakpoints Properties
Filter
element.style {
@media (min-width: 601px) {
.box {
background-color: blue;
task7.css? ...99657020:26
.box {
width: 100%;
height: 200px;
background-color: blue;
color: white;
text-align: center;
line-height: 200px;
font-size: 20px;
Console Issues


127.0.0.1:5500/task7.html
Dimensions: Responsive 580 x 549 100% No throttling 'Save-Data': default
Resize the screen to see color changes
Elements
<!DOCTYPE html>
<html lang="en">
<head>
<body>
Resize the screen to see color changes
<!-- Code injected by live-server -->
<script>
</body>
</html>
html body div.box
Styles Computed Layout Event Listeners DOM Breakpoints Properties
Filter
element.style {
@media (min-width: 400px) and (max-width: 600px) {
.box {
background-color: red;
task7.css? ...99657020:21
.box {
width: 100%;
height: 200px;
background-color: blue;
color: white;
text-align: center;
line-height: 200px;
font-size: 20px;
Console Issues


127.0.0.1:5500/task7.html
Dimensions: Responsive 368 x 549 100% No throttling 'Save-Data': default
Resize the screen to see color changes
Elements
<!DOCTYPE html>
<html lang="en">
<head> ... </head>
<body>
Resize the screen to see color changes
<!-- Code injected by live-server -->
<script> ... </script>
</body>
</html>
html body div.box
Styles Computed Layout Event Listeners DOM Breakpoints Properties
Filter
element.style {
}
@media (min-width: 300px) and (max-width: 399px) {
.box {
background-color: pink;
}
}
task7.css?_=99657020:16
.box {
width: 100%;
height: 200px;
background-color: blue;
color: white;
text-align: center;
line-height: 200px;
font-size: 20px;
}
Console Issues

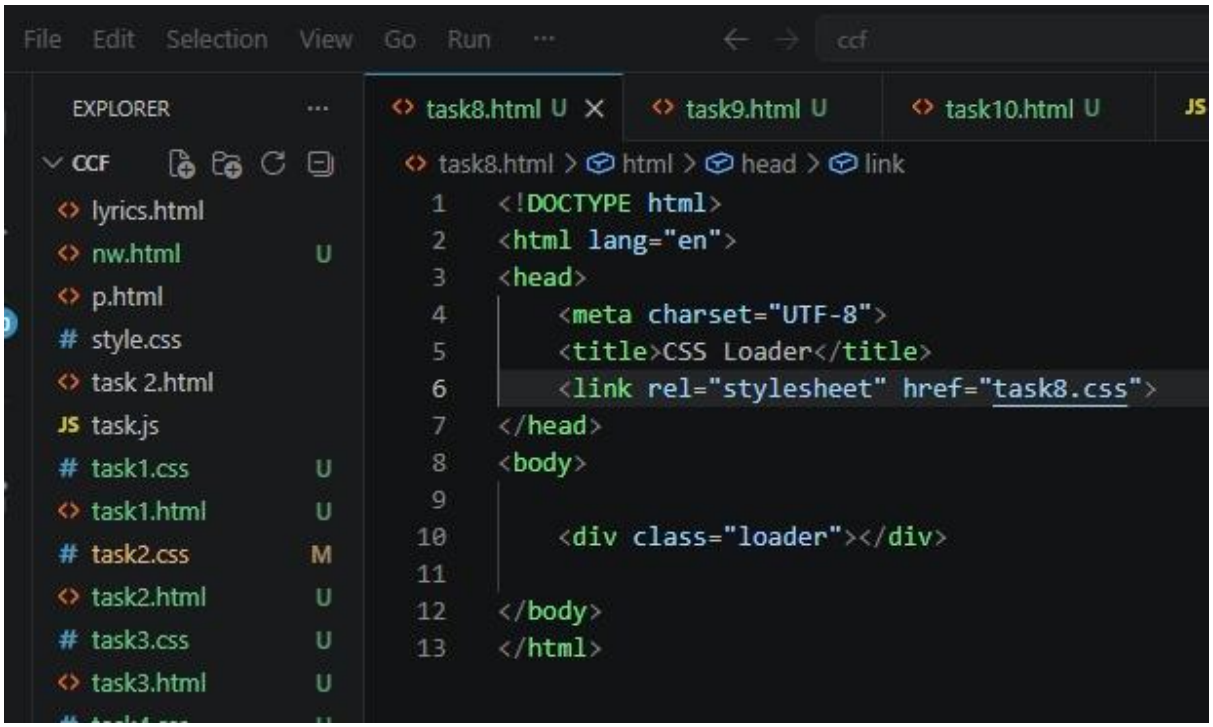

127.0.0.1:5500/task7.html
Dimensions: Responsive 262 x 549 100% No throttling 'Save-Data': default
Resize the screen to see color
Elements
<!DOCTYPE html>
<html lang="en">
<head> ... </head>
<body>
Resize the screen to see color changes
<!-- Code injected by live-server -->
<script> ... </script>
</body>
</html>
html body div.box
Styles Computed Layout Event Listeners DOM Breakpoints Properties
Filter
element.style {
}
@media (max-width: 299px) {
.box {
background-color: green;
}
}
task7.css?_=99657020:11
.box {
width: 100%;
height: 200px;
background-color: blue;
color: white;
text-align: center;
line-height: 200px;
font-size: 20px;
}
Console Issues

Practice Set -8

Qs: Create a simple loader using CSS

- Step1: create a div with circular shape & a thick border from one end (top/bottom/left/right) .
- Step2: To make it spin create an animation which transforms it from 0deg to 360deg.
- Step3: Add the animation property to the loader with infinite duration. Source

SOURCECODE:-



CSS:-

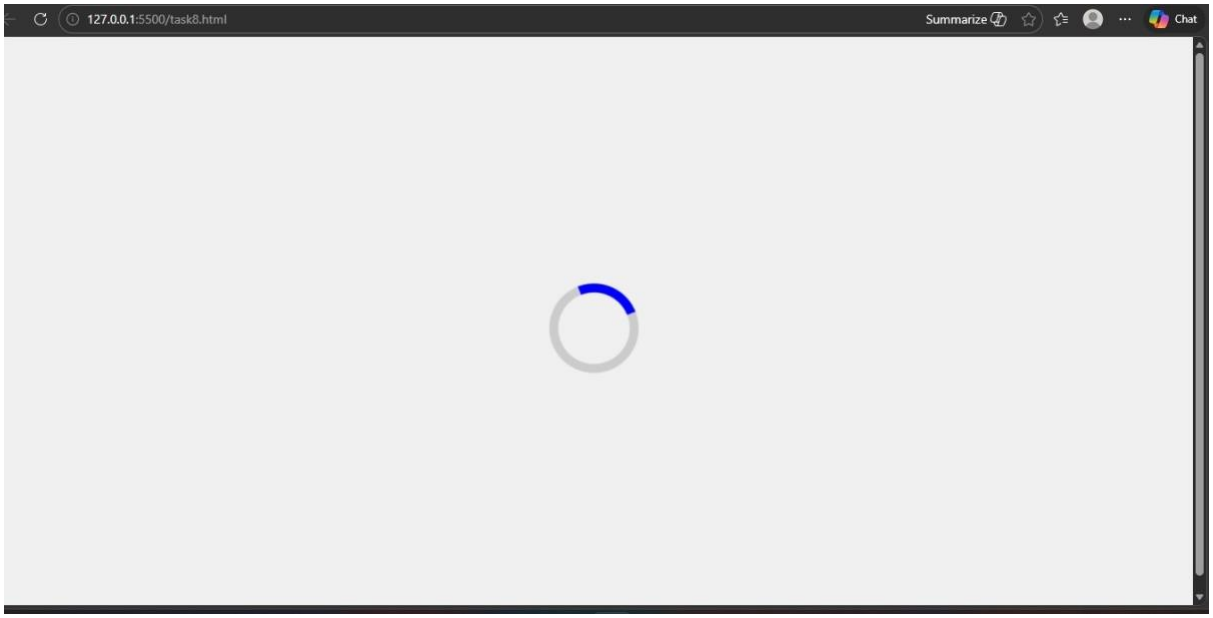
Go Run ··· ccf

task9.html U | task10.html U | task10.js U

task8.css > ...

```
body {
    display: flex;
    justify-content: center;
    align-items: center;
    height: 100vh;
    background-color: #f0f0f0;
}


.loader {
    width: 80px;
    height: 80px;
    border: 10px solid #ccc;
    border-top: 10px solid blue;
    border-radius: 50%;

    animation: spin 1s linear infinite;
}


@keyframes spin {
    from {
        transform: rotate(0deg);
    }
    to {
        transform: rotate(360deg);
    }
}
```

Output :



.

PRACTICE SET 9.

: Create a simpleHTMLpage wita heading and a paragraph and a button. Link a separate JavaScript file

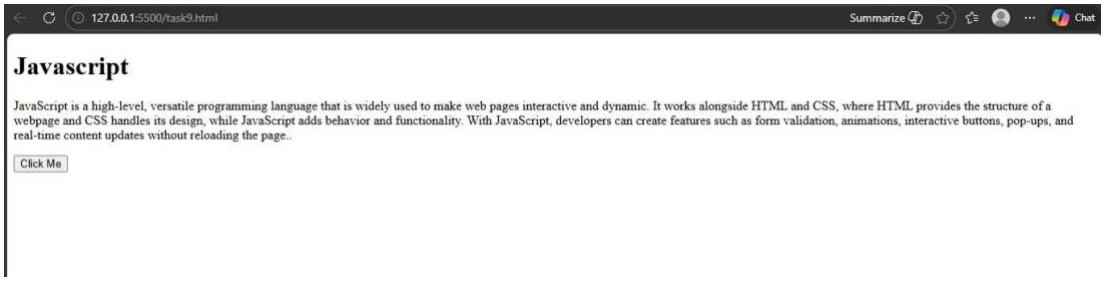
SOURCECODE
HTML :-

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Simple Page</title>
</head>
<body>

    <h1 id="heading">Javascript</h1>
    <p id="para">JavaScript is a high-level, versatile programming language that is widely used to
        make web pages interactive and dynamic.It works alongside HTML and CSS, where HTML provides
        the structure of a webpage and CSS handles its design, while JavaScript adds behavior and functionality.
        With JavaScript, developers can create features such as form validation, animations, interactive buttons,
        pop-ups, and real-time content updates without reloading the page..</p>
    <button onclick="changeText()">Click Me</button>

    <!-- Link to external JavaScript -->
    <script src="task9.js"></script>

</body>
</html>
```

Js :-

```
function changeText() {
    document.getElementById("heading").innerHTML = "Hello, You Clicked the Button!";
    document.getElementById("para").innerHTML = "The text has been changed using JavaScript.";
}
```

Output :-



PRACTICE SET :10

: Create a simple webpage that displays the name of an animal at the top of the page. Add an interactive feature so that when the mouse pointer moves over the animal's name, the name changes to a fun nickname. When the mouse pointer moves away, the name should change back to the original animal's name.

To make the page more interactive, also use set Timeout to display a small message after a few seconds. For example, after 3 or 4 seconds, a message such as

"Welcome to the Animal Page!"

Source code :-

HTML :-

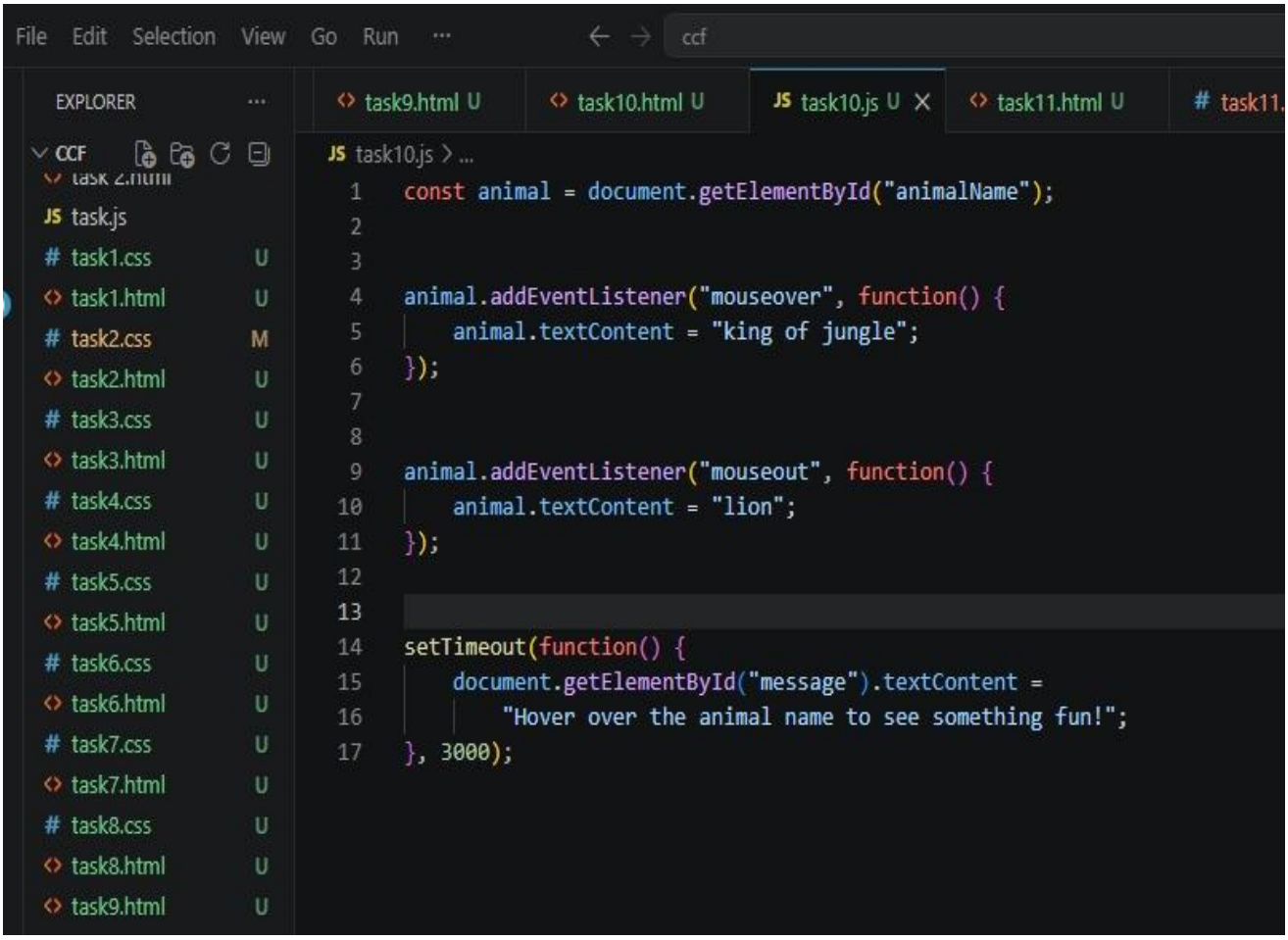
Go Run ··· ccf

task9.html U | task10.html U | task10.js U | task11.html U | task11.css 6, U

task10.html > html > body > h1#animalName

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>task10</title>
</head>
<body style="text-align:center; font-family:Arial;">

    <!-- Animal Name -->
    <h1 id="animalName">lion</h1>

    <!-- Message -->
    <p id="message"></p>

    <!-- Link JavaScript -->
    <script src="task10.js"></script>

</body>
</html>
```

Js :-



PRACTICE SET :11

: Create a webpage with a heading and enough content so the page can scroll.
Add two interactive features using JavaScript.

1. Dark Mode
   - Add a button to toggle dark and light mode.
   - When clicked:

O Background and text colours change. O Button text changes:
Dark Mode ↔ ⚙ Light Mode.

2. Smart Scroll-to-Top
   - Add a "↑ Back to Top" button at the bottom corner of the page.
   - Button appears only when the user scrolls down.
   - Clicking the button smoothly scrolls the page to the top.

Source code

HTML:

Go Run ...

task9.html U | task10.html U | task10.js U | task11.html U × | task11.css 6, U | task11.js U | task9.js U

task11.html > html > body > div.glass > div.content > p

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>task11</title>
    <link rel="stylesheet" href="task11.css">
</head>
<body>

<div class="glass">
    <div class="navbar">
    <h1 id=""toggle txt"> climate</h1>
    <label class="theme-toggle">
        <input type="checkbox" id="toggle"/>
        <div class="toggle bg">
            <div class="toggle-circle">
                <div class="sun"></div>
                <div class="moon"></div>
            </div>
            <div class="stars"></div>
        </div>
    </label>
</div>
<div class="content">
    <p>Climate refers to the long-term pattern of weather conditions in a particular region over an extended p
 characteristics of a place. Unlike weather, which can change from day to day or even hour to hour, climate re
  provides a broader understanding of how the atmosphere behaves over time. The study of climate is important
and researchers understand natural processes, predict future environmental conditions, and plan for sustainabl
has a direct impact on human life, influencing agriculture, water availability, housing, clothing, and even cu
with warm climates may support crops like rice and coconut, while colder regions are suitable for crops like w
influence the climate of a region, making each place unique in its environmental conditions. One of the most
```

Ln 30, Col 5 Spaces: 4 UTF-8 LF

```
Go  Run  ···                ccf
task9.html U   task10.html U   task10.js U   task11.html U ×   task11.css 6, U   task11.js U   task9.js U   task8.css U
task11.html > html > body > div.glass > div.content > p
2   <html lang="en">
9   <body>
11  <div class="glass">
25  <div class="content">
26      <p>Climate refers to the long-term pattern of weather conditions in a particular region over an extended period,usually 3
39   side of a mountain may receive heavy rainfall, while the leeward side remains dry, creating a rain shadow region. Wind patte
40    by carrying heat and moisture across different regions of the Eart
41  The world's climates are generally classified into several major types based on temperature and precipitation patterns. The t
42   found near the equator, is characterized by high temperatures and heavy rainfall throughout the year, supporting dense fores
43    variety of plant and animal life. The dry or arid climate, which includes deserts, experiences very little rainfall and has
44     with organisms adapted to survive in harsh conditions. The temperate climate, found in regions between the tropics and the
45     has moderate temperatures and distinct seasons, making it highly suitable for agriculture and human habitation. The polar
46   near the Earth's poles, is extremely cold, with ice and snow covering the land for most of the year and supporting only limi
47    Each of these climate types plays a vital role in shaping ecosystems and biodiversity across the globe.
48  In recent years, climate change has become one of the most significant environmental challenges facing the world.
49  Climate change refers to long-term alterations in temperature and weather patterns, largely caused by human activities
50   such as the burning of fossil fuels, deforestation, industrialization, and increased use of vehicles. These activities
51    release greenhouse gases like carbon dioxide and methane into the atmosphere, which trap heat and lead to global warming.
52     As a result, the Earth's average temperature is rising, causing glaciers and polar ice caps to melt, leading to rising sea
53     levels and increased risk of coastal flooding. Climate change also contributes to more frequent and intense extreme weathe
54     events such as hurricanes, droughts, heatwaves, and heavy rainfall. These changes not only affect natural ecosystems but a
55      serious threats to human health, food security, and water resources.
56  Climate is extremely important for sustaining life on Earth, as it influences the distribution of plants, animals, and human
57  It determines the types of crops that can be grown in a region, the availability of water, and the overall living conditions
58   A stable climate ensures balanced ecosystems and supports biodiversity, while drastic changes can disrupt natural systems an
59   environmental degradation. Therefore, understanding climate and taking steps to protect it are essential for ensuring a sust
60  SSSssEfforts such as reducing greenhouse gas emissions, conserving forests, using renewable energy sources, and promoting env
61  practices can help mitigate the impacts of climate change. In conclusion, climate is a fundamental aspect of the Earth's syst
62  every living organism, and preserving it is crucial for the well-being of present and future generations.Climate refers to th
63  pattern of weather conditions in a particular region over an extended period, usually spanning several decades. It includes e
64  temperature, rainfall, humidity, wind speed and direction, and seasonal changes that together describe the average atmospheri
65  conditions. The study of climate is essential because it helps us understand how the Earth's systems function, how different
```

```
Go  Run  ···                ccf
task9.html U   task10.html U   task10.js U   task
task11.html > html > body > div.glass > div.content > p
2   <html lang="en">
9   <body>
11  <div class="glass">
25  <div class="content">
75  </p>
76  </div>
77
78  <button id="scrolltoup">
79  <svg
80      width="100%"
81      height="100%"
82      viewBox="0 0 24 24"
83      fill="none"
84      xmlns="http://www.w3.org/2000/svg"
85
86      <path
87          d="M12 4L12 20M12 4L5 11M12 4L9 11"
88          stroke="currentColor"
89          stroke-width="2"
90          stroke-linecap="round"
91          stroke-linejoin="round"
92          />
93  </svg>
94  </button>
95  </div>
96  <script src=" task11.js"></script>
97  </body>
98  </html>
```

CSS :-

Go Run ··· ccf

task9.html U | task10.html U | JS task10.js

task11.css > body

```
html {
    scroll-behavior: smooth;
}
body {
    margin: 0;
    background:var(--bg);
    color:var(--tx);
    transition: 0.4s;
    height: 100vh;
    dispaly:grid;
}
.glass {
    max-width:1000px;
    margin:auto;
}
:root{
    --bg: #7bf6a4;
    --tx: black;
}
.dark{
    --bg: #0b0f1a;
    --tx: white;
}
.navbar{
    display: flex;
    flex-direction: row;
    justify-content: space-between;
    align-items: center;
    width: 100%;
    height: 60px;
}
.theme-toggle{
```

Go Run ··· ccf

task9.html U | task10.html U | JS task10.js U | task11.html U

task11.css > .toggle-circle

```
.navbar{
}
.theme-toggle{
    width: 80px;
    height: 40px;
    cursor:pointer;
}
.theme-toggle input{
    display: none;
}
.toggle-bag{
    width: 100%;
    height: 100%;
    background: linear-gradient(to right, #87ceeb, #e0f6ff);
    border-radius: 40px;
    position: relative;
    overflow: hidden;
}
.toggle-circle{
    width:40px;
    height: 40px;
    background-color: #fff;
    border: radius 50%;
    position: absolute;
    top: 3px;
    left: 3px;
    display: grid;
    place-items: center;
    transform: 0.4s;
}
.sun{
    width: 20px;
```

Go Run ··· ccf

task9.html U | task10.html U | task10.js U | task11.html U

task11.css > .toggle-circle

```
.sun{
    background-color: gold;
    border-radius: 50%;
    position: absolute;
    transition:0.4s;
}
.moon{
    width:20px;
    height: 20px;
    background: #ddd;
    border-radius: 50%;
    position: absolute;
    transition: 0.4s;
    clip-path: circle(50% at 30% 50%);
    opacity: 0;
}
input:checked + .toggle-bg{
    background: linear-gradient(to right,$0b0f1a,#1a1f2e);
}
input:checked + .toggle-bg .toggle-circle{
    transform: translate(40px);
}
input:checked + .toggle-bg .sun{
    opacity:0;
}
input:checked +.toggle-bg .moon{
    opacity: 1;
}
.stars{
    position:absolute;
    width:100%;
```

Go Run ··· ccf

task9.html U | task10.html U | task10.js U | task11.html U | # task

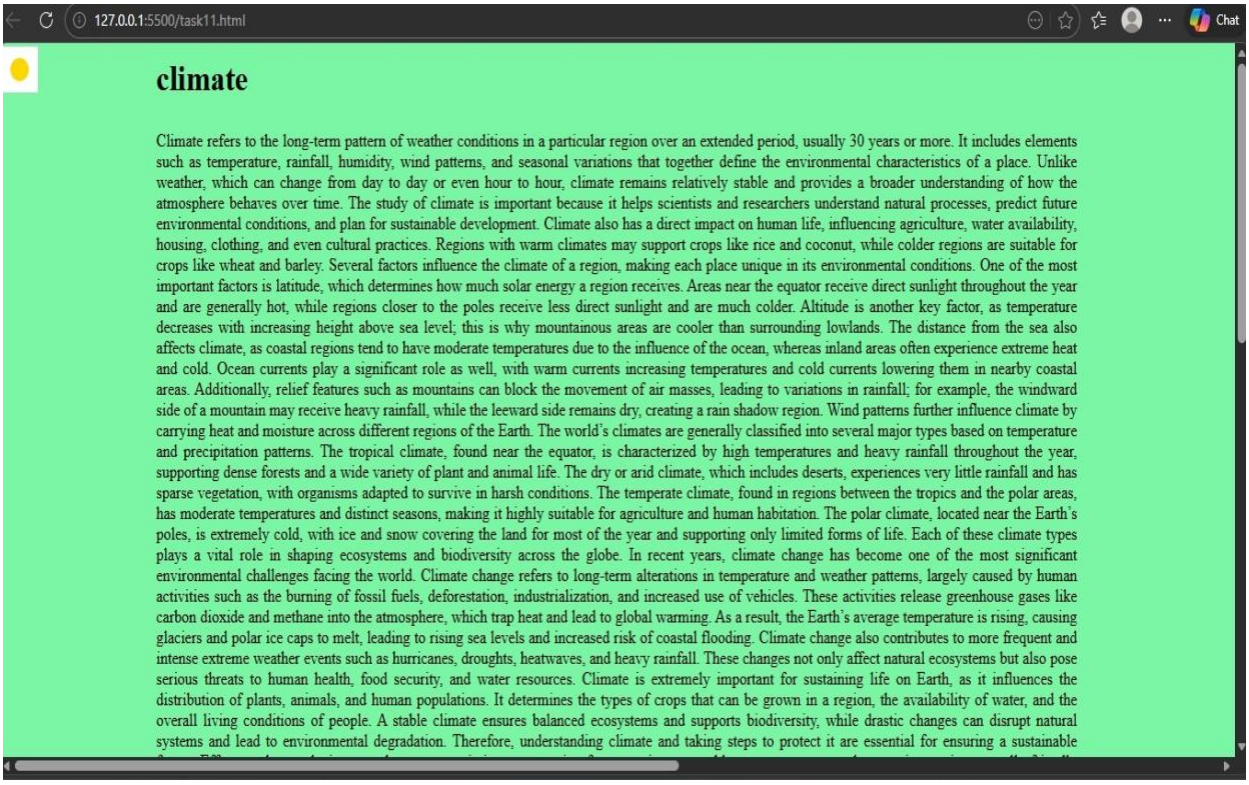
task11.css > .toggle-circle

```
.stars{
    width:100%;
    height:100%;
    background:
        radial-gradient(2px 2px at 20% 30%, white,transparent),
        radial-gradient(2px 2px at 60% 70%, white,transparent),
        radial-gradient(1.5px 1.5px at 80% 20%, white,transparent),
    opacity: 0;
   transform: 1;
}
input:checked + .toggle-bg .stars{
    opacity: 1;
}
p{
    text-align: justify;
}
#scrolltoup{
    width: 30px;
    height: 30px;
    border-radius: 50%;
    position: fixed;
    bottom: 20px;
    right: 20px;
    display: none;
    cursor:pointer;
}
#scrolltoup:hover{
    transform: scale(1,1);
}
```

JS:-

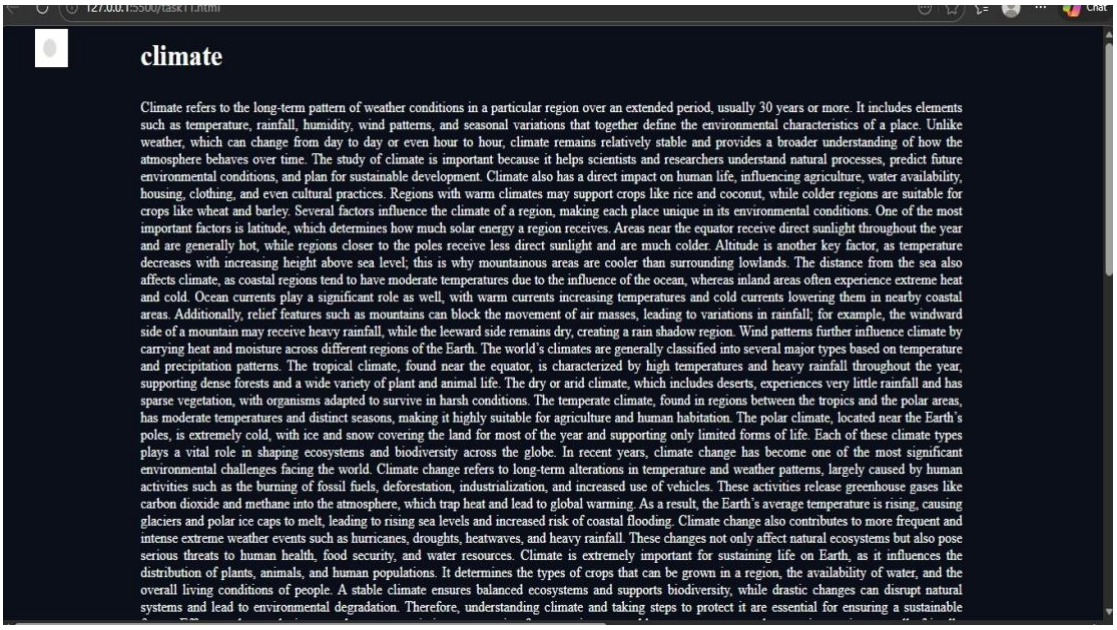
```js
const toggle = document.getElementById("toggle");
toggle.addEventListener("change",() => {
    document.body.classList.toggle("dark");
});

const btn=document.getElementById("scrolltoup");
window.addEventListener("scroll", () => {
    if (window.scrollY > 200) {
        btn.style.display = "block";
    } else {
        btn.style.display = "none";
    }
});
btn.addEventListener("click", () => {
    window.scrollTo({
        top: 0,
        behavior: "smooth"
    });
});
```

OUTPUT :-



climate

Climate refers to the long-term pattern of weather conditions in a particular region over an extended period, usually 30 years or more. It includes elements such as temperature, rainfall, humidity, wind patterns, and seasonal variations that together define the environmental characteristics of a place. Unlike weather, which can change from day to day or even hour to hour, climate remains relatively stable and provides a broader understanding of how the atmosphere behaves over time. The study of climate is important because it helps scientists and researchers understand natural processes, predict future environmental conditions, and plan for sustainable development. Climate also has a direct impact on human life, influencing agriculture, water availability, housing, clothing, and even cultural practices. Regions with warm climates may support crops like rice and coconut, while colder regions are suitable for crops like wheat and barley. Several factors influence the climate of a region, making each place unique in its environmental conditions. One of the most important factors is latitude, which determines how much solar energy a region receives. Areas near the equator receive direct sunlight throughout the year and are generally hot, while regions closer to the poles receive less direct sunlight and are much colder. Altitude is another key factor, as temperature decreases with increasing height above sea level; this is why mountainous areas are cooler than surrounding lowlands. The distance from the sea also affects climate, as coastal regions tend to have moderate temperatures due to the influence of the ocean, whereas inland areas often experience extreme heat and cold. Ocean currents play a significant role as well, with warm currents increasing temperatures and cold currents lowering them in nearby coastal areas. Additionally, relief features such as mountains can block the movement of air masses, leading to variations in rainfall; for example, the windward side of a mountain may receive heavy rainfall, while the leeward side remains dry, creating a rain shadow region. Wind patterns further influence climate by carrying heat and moisture across different regions of the Earth. The world's climates are generally classified into several major types based on temperature and precipitation patterns. The tropical climate, found near the equator, is characterized by high temperatures and heavy rainfall throughout the year, supporting dense forests and a wide variety of plant and animal life. The dry or arid climate, which includes deserts, experiences very little rainfall and has sparse vegetation, with organisms adapted to survive in harsh conditions. The temperate climate, found in regions between the tropics and the polar areas, has moderate temperatures and distinct seasons, making it highly suitable for agriculture and human habitation. The polar climate, located near the Earth's poles, is extremely cold, with ice and snow covering the land for most of the year and supporting only limited forms of life. Each of these climate types plays a vital role in shaping ecosystems and biodiversity across the globe. In recent years, climate change has become one of the most significant environmental challenges facing the world. Climate change refers to long-term alterations in temperature and weather patterns, largely caused by human activities such as the burning of fossil fuels, deforestation, industrialization, and increased use of vehicles. These activities release greenhouse gases like carbon dioxide and methane into the atmosphere, which trap heat and lead to global warming. As a result, the Earth's average temperature is rising, causing glaciers and polar ice caps to melt, leading to rising sea levels and increased risk of coastal flooding. Climate change also contributes to more frequent and intense extreme weather events such as hurricanes, droughts, heatwaves, and heavy rainfall. These changes not only affect natural ecosystems but also pose serious threats to human health, food security, and water resources. Climate is extremely important for sustaining life on Earth, as it influences the distribution of plants, animals, and human populations. It determines the types of crops that can be grown in a region, the availability of water, and the overall living conditions of people. A stable climate ensures balanced ecosystems and supports biodiversity, while drastic changes can disrupt natural systems and lead to environmental degradation. Therefore, understanding climate and taking steps to protect it are essential for ensuring a sustainable



climate

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127.0.0.1:5500/task11.html

in regulating temperatures and influencing rainfall patterns in nearby coastal regions. Relief features such as mountains can act as barriers to wind, causing heavy rainfall on one side and dry conditions on the other, known as the rain shadow effect. Additionally, prevailing winds carry moisture and heat across the globe, further shaping regional climates. Climate classification helps scientists group different regions of the world based on similar climatic conditions. The tropical climate, found near the equator, is characterized by consistently high temperatures and heavy rainfall, supporting dense forests and rich biodiversity. The dry or arid climate includes deserts and semi-arid regions where rainfall is scarce, temperatures can be extreme, and vegetation is limited. The temperate climate, located between the tropics and polar regions, experiences moderate temperatures and distinct seasons, making it ideal for agriculture and human habitation. The continental climate, found in the interiors of continents, is marked by hot summers and cold winters with significant temperature variations. The polar climate, located near the poles, is extremely cold with ice-covered landscapes and limited life forms. In addition to these major types, there are also specialized climates such as Mediterranean, tundra, and monsoon climates, each with unique characteristics and ecological importance. Climate also has a strong connection with ecosystems and biodiversity. Different climatic conditions support different types of vegetation and wildlife. For example, tropical rainforests are home to a vast number of species due to warm temperatures and abundant rainfall, while deserts support specially adapted plants and animals that can survive with minimal water. Grasslands, forests, tundra, and aquatic ecosystems all depend on specific climate conditions to thrive. Any significant change in climate can disrupt these ecosystems, leading to loss of biodiversity and imbalance in natural systems. This shows how closely climate is linked to the health of the environment and the survival of living organisms. In recent decades, the issue of climate change has gained global attention due to its widespread impacts. Climate change refers to long-term changes in temperature and weather patterns, largely caused by human activities such as burning fossil fuels, deforestation, industrial emissions, and urbanization. These activities release greenhouse gases like carbon dioxide, methane, and nitrous oxide into the atmosphere, which trap heat and cause the Earth's temperature to rise. This phenomenon, known as global warming, has led to melting glaciers, rising sea levels, and increased frequency of extreme weather events such as floods, droughts, cyclones, and heatwaves. Climate change also affects agriculture by altering rainfall patterns and growing seasons, which can lead to food shortages and economic challenges. Furthermore, it poses serious risks to human health, including heat-related illnesses and the spread of diseases. Efforts to address climate change involve both mitigation and adaptation strategies. Mitigation focuses on reducing the causes of climate change by lowering greenhouse gas emissions through the use of renewable energy sources such as solar, wind, and hydroelectric power, improving energy efficiency, and promoting sustainable transportation. Reforestation and afforestation are also important measures, as trees absorb carbon dioxide and help regulate the climate. Adaptation, on the other hand, involves adjusting to the impacts of climate change by developing resilient infrastructure, improving water management, and adopting climate-smart agricultural practices. Governments, organizations, and individuals all have a role to play in protecting the climate by making environmentally responsible choices. Climate is also important for economic development and planning. Many industries, such as agriculture, tourism, and fisheries, depend heavily on climate conditions. For example, tourism in hill stations, beaches, and snowy regions is influenced by seasonal climate patterns. Similarly, farmers rely on predictable rainfall and temperature conditions for crop production. Any disruption in climate can therefore have significant economic consequences. Understanding climate helps in disaster management as well, allowing authorities to prepare for and respond to natural disasters like hurricanes, floods, and droughts more effectively. In conclusion, climate is a fundamental aspect of the Earth's system that influences natural environments, human activities, and the overall balance of life on the planet. It is shaped by a complex interaction of natural factors and is now increasingly affected by human actions. The growing concern over climate change highlights the need for greater awareness and collective action to protect the environment. By adopting sustainable practices, reducing pollution, conserving natural resources, and supporting global efforts to combat climate change, it is possible to preserve the Earth's climate for future generations. Understanding climate in detail not only helps us appreciate the natural world but also encourages us to take responsibility for maintaining the delicate balance that sustains life on Earth.